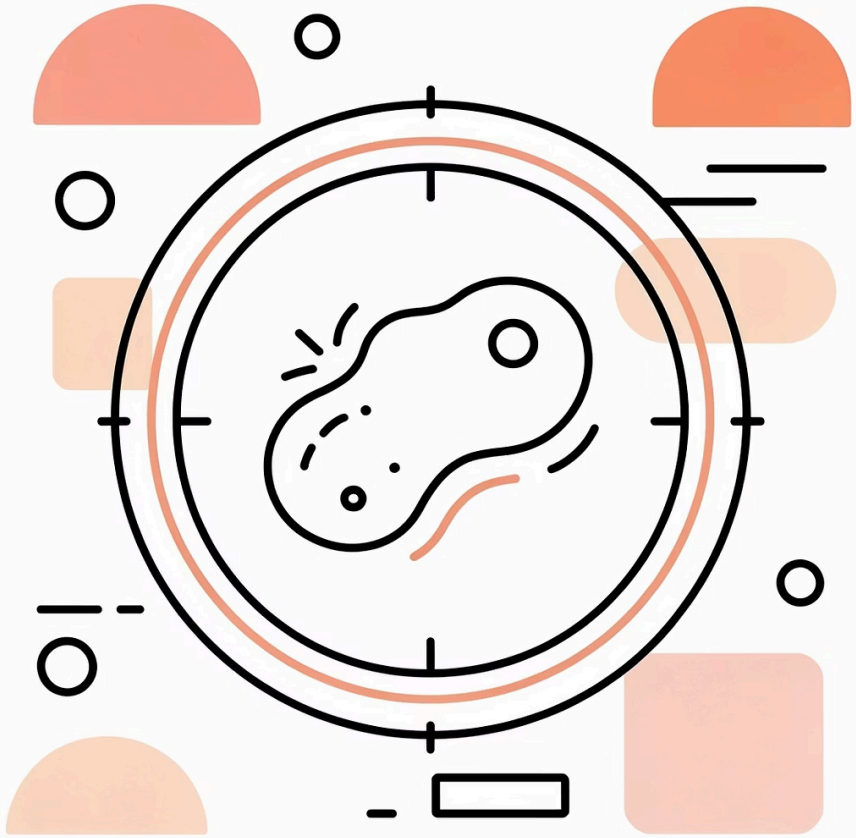


Melanoma Detection with MobileNetV2

CP4210 Computer Vision · ISIC 2018 Dataset

EDUCATIONAL DEMO ONLY — NOT A MEDICAL DEVICE

Why Early Detection Matters

Skin cancer is the **most common cancer globally**, with melanoma accounting for the majority of skin cancer deaths despite representing only a small fraction of cases. When detected at Stage I, the 5-year survival rate exceeds 98%. By Stage IV, that drops to under 25%.

AI-assisted dermoscopy offers a scalable path to earlier detection — flagging suspicious lesions for clinical review *before* a biopsy is required. This project demonstrates how modern computer vision techniques can be applied to dermoscopic imagery to build a practical, interpretable screening tool.

Key Motivation

- Melanoma is deadly but treatable when caught early
- Dermoscopy improves diagnostic accuracy over naked-eye inspection
- CNN-based models can approach dermatologist-level performance
- Explainability (Grad-CAM) builds clinical trust

Problem Definition

This project frames skin lesion classification as a **binary task**: distinguishing melanoma from all other lesion types. The primary challenge is not accuracy alone — it is building a model that catches the dangerous positives despite a heavily skewed dataset.

Binary Classification Task

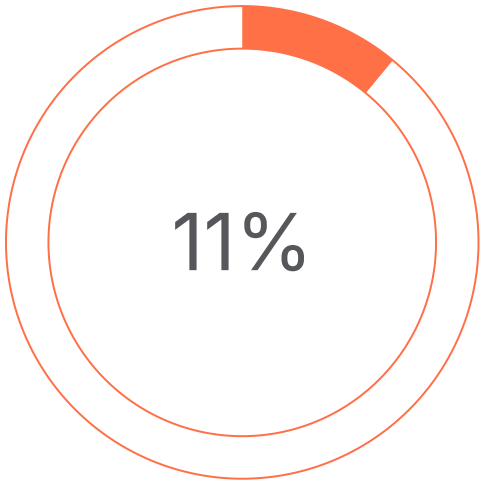
Melanoma (positive class) vs. Non-melanoma (negative class). Labels derived from ISIC 2018 Task 3 multi-class annotations collapsed to a binary schema.

Severe Class Imbalance

Only ~11% of images are melanoma-positive. A naive classifier predicting "Non-melanoma" always would achieve ~89% accuracy — making recall-focused metrics essential.

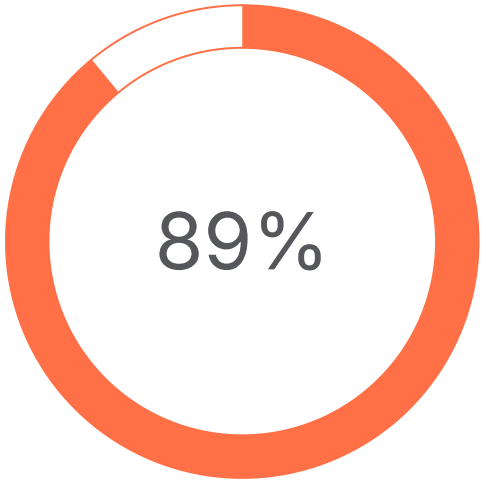
Evaluation Priority

F1 score and Recall are the primary metrics. Missing a true melanoma (false negative) carries a much higher clinical cost than a false positive.



Melanoma Positive

Share of positive class in the ISIC 2018 dataset



Non-Melanoma

Dominant negative class driving imbalance

Dataset Overview: ISIC 2018 Task 3

The **International Skin Imaging Collaboration (ISIC) 2018 Task 3** benchmark provides approximately 10,000 dermoscopic images across seven diagnostic categories. For this project, all non-melanoma classes are merged into a single negative class to create a clean binary classification scenario.

~10,000 Images

Total dermoscopy images from ISIC 2018 Task 3, professionally labeled by dermatologists

70 / 15 / 15 Split

Stratified train / validation / test partition, preserving class ratios across all splits

Binary Labels

Multi-class annotations collapsed: Melanoma = 1, all other lesion types = 0

Data Augmentation

Horizontal flip, random rotation, and color jitter applied during training to improve generalization

Methodology: Transfer Learning with MobileNetV2

MobileNetV2 was selected for its favorable balance of parameter efficiency and accuracy. Pretrained on ImageNet (1.2M images, 1000 classes), the backbone already encodes rich visual features — textures, edges, color gradients — directly applicable to dermoscopy.

The final classification head is replaced with a **2-class linear layer**, and the entire network is fine-tuned end-to-end. This approach requires far less data and compute than training from scratch, while leveraging deep feature representations learned at scale.

Transfer learning is especially advantageous in medical imaging, where large labeled datasets are scarce and annotation is expensive.

Load Pretrained Backbone

MobileNetV2 weights from ImageNet — fine-tuned end-to-end from epoch 1 with AdamW at lr=3e-4, no layer freezing

Replace Classifier Head

Swap final layer: 1000-class softmax → 2-class linear projection

Fine-Tune End-to-End

All layers trained jointly with AdamW — low learning rate preserves pretrained features while adapting to ISIC data

Methodology: Training Details

Every hyperparameter choice was made deliberately to address the core challenges of **class imbalance**, **limited training time**, and **stable convergence** on a relatively small medical dataset.

Optimizer: AdamW

Learning rate `3e-4`, weight decay `1e-4`. AdamW decouples L2 regularization from gradient updates, improving generalization over vanilla Adam.

Weighted Cross-Entropy Loss

Class weights inversely proportional to frequency. Upweights the melanoma class to penalize false negatives more heavily during training.

Mixed-Precision (AMP)

Automatic Mixed Precision reduces memory footprint and speeds up forward/backward passes on compatible GPUs without sacrificing numerical stability.

Best Checkpoint by F1

Model trained for 5 epochs. Best checkpoint selected by validation F1 score — not accuracy — to prioritize recall on the positive class.

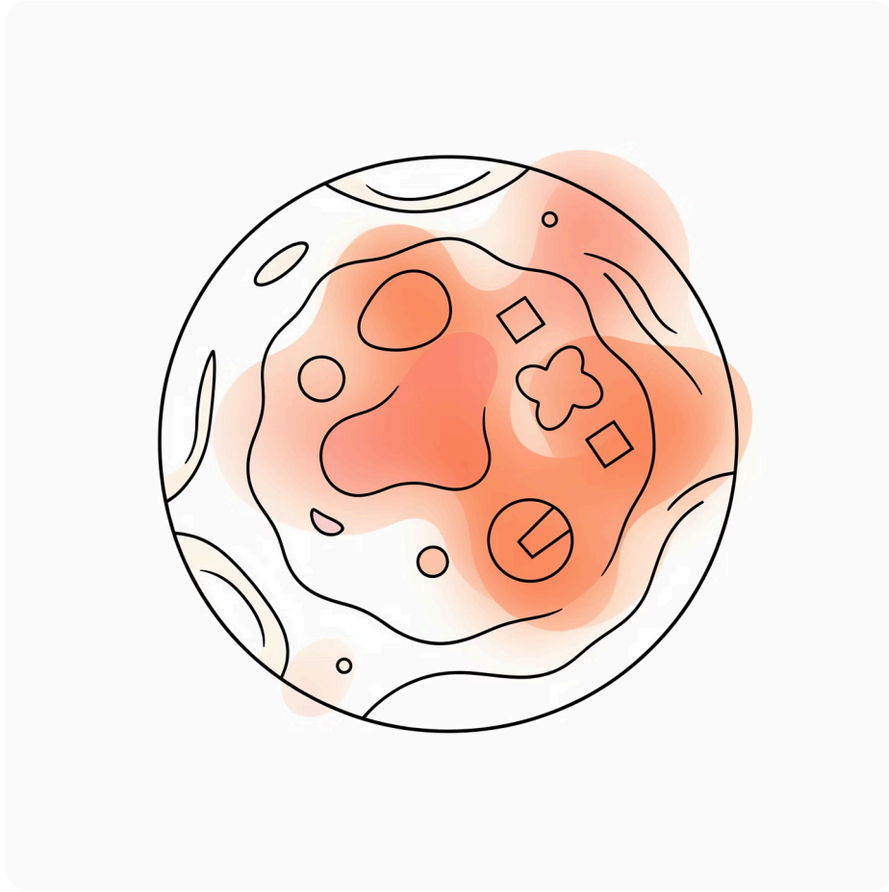
Methodology: Explainability via Grad-CAM

Gradient-Weighted Class Activation Mapping

Grad-CAM computes the gradient of the predicted class score with respect to the final convolutional feature maps. These gradients are global-average-pooled to produce per-channel importance weights, which are then linearly combined with the feature maps to generate a coarse localization heatmap.

Applied to the **last convolutional layer of MobileNetV2**, Grad-CAM highlights the spatial regions that most influenced the model's prediction — without requiring any architectural changes or retraining.

Clinical value: Helps dermatologists verify whether the model is attending to medically meaningful structures (e.g., irregular borders, color asymmetry) rather than background artifacts.



No Retraining Required

Post-hoc, applied at inference time

Spatial Localization

Produces pixel-level importance map

Results: Test Set Performance

After 5 training epochs, the model achieves **89.9% test accuracy** — near the naive baseline — but more importantly, it captures **69.5% of all true melanoma cases** (recall), demonstrating genuine discriminative ability beyond trivial majority-class prediction.

89.9%

Accuracy

69.5%

Recall

53.5%

Precision

60.4%

F1 Score

Confusion Matrix

	Predicted: Neg	**Predicted: Pos**
Actual: Neg	TN = 1,235	FP = 101
Actual: Pos	FN = 51	TP = 116

The model correctly identifies 116 of 167 melanoma cases. The 51 false negatives represent missed melanomas — the primary target for future improvement. The 101 false positives lead to unnecessary follow-up but carry lower clinical risk.

Results: ROC & Precision-Recall Curves

ROC Curve

The ROC curve plots the true positive rate against the false positive rate across all classification thresholds. A model with no discriminative ability would follow the diagonal. **Area Under the Curve (AUC)** summarizes overall separability — the higher the AUC, the better the model distinguishes melanoma from non-melanoma regardless of threshold choice.

Strong AUC despite class imbalance indicates the model has learned meaningful features, not just the majority-class distribution.

Precision-Recall Curve

The PR curve is especially informative under class imbalance. It plots precision against recall across thresholds, revealing the trade-off between catching more melanomas (higher recall) and generating more false alarms (lower precision). The **area under the PR curve (AUPRC)** is a more honest metric than AUC-ROC when positives are rare.

Threshold tuning in the Streamlit dashboard allows practitioners to shift the operating point along this curve to match clinical requirements.

0.925

AUC-ROC

Strong separability — well above the 0.5 random baseline

0.670

AUPRC

Honest metric under class imbalance — reflects true positive precision

Interactive Streamlit Dashboard

The project is accompanied by a fully interactive **Streamlit web application** that exposes every diagnostic layer of the model — designed for students, researchers, and clinicians who want to explore model behavior beyond raw metrics.

Threshold Tuning

Interactively adjust the classification threshold and watch precision, recall, and F1 update in real time

Confusion Matrix

Live confusion matrix updates as threshold changes, making the precision-recall trade-off tangible

Error Explorer

Gallery of false positives and false negatives — drill into individual misclassified cases

Grad-CAM Overlay

Visualize the activation heatmap for any image — uploaded live or selected from the test set

Live Inference

Upload any dermoscopy image and receive a real-time prediction with confidence score and Grad-CAM visualization

Calibration Plot

Reliability diagram showing how well model confidence aligns with empirical accuracy across probability bins

Limitations & Future Work

Current Limitations

→ Domain Shift

Model trained on clinical dermoscopy images; performance degrades on smartphone photos due to lighting, focus, and perspective differences

→ Limited Training Duration

Only 5 epochs — longer training with learning rate scheduling would likely improve recall further and reduce overfitting risk

→ Binary Scope Only

Collapsing 7 ISIC classes to binary discards clinically relevant distinctions between benign lesion subtypes

Promising Research Directions

🔬 Stronger Architectures

EfficientNet-B4 or ResNet-50 with attention mechanisms for higher baseline capacity

📊 Multi-Class Classification

Restore full 7-class ISIC taxonomy with class-aware sampling strategies

🤝 Ensemble Methods

Combine MobileNetV2, EfficientNet, and ViT predictions via soft voting or stacking

📱 Domain Adaptation

Fine-tune on mixed dermoscopy + smartphone datasets for real-world robustness

Conclusion

This project demonstrates a complete, end-to-end computer vision pipeline for a clinically meaningful task — from raw dermoscopy images to an interactive, explainable diagnostic dashboard.

Strong Baseline Performance

MobileNetV2 achieves **89.9% accuracy** and **69.5% melanoma recall** in just 5 training epochs, demonstrating the power of transfer learning on limited medical data

Interpretable Predictions

Grad-CAM heatmaps surface the spatial evidence behind each decision, grounding model outputs in clinically verifiable visual features rather than opaque scores

Core CV Techniques Demonstrated

Transfer learning, stratified data splitting, data augmentation, weighted loss for class imbalance, mixed-precision training, and post-hoc explainability

⚠️ **Educational Use Only:** This system is a research and learning demonstration built for CP4210. It is not validated for clinical use, does not constitute medical advice, and should not be used as a substitute for professional dermatological examination.